

Article

A Survey on Enterprise Resource Planning and Building Information Modeling Integration: A Construction 4.0 Perspective

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Abstract: Managing the fragmented nature of information generated on construction sites and firms is the need of the hour. Building Information Modeling (BIM) and Enterprise Resource Planning (ERP) are the most prevalent information systems (ISs) used in the construction industry, with varied capabilities but also with significant overlap. It is essential to analyze the current state of Common Data Environments (CDEs) that support both BIM and ERP as separate systems, as well as those that facilitate their integration. This analysis will ultimately help establish a practical framework for implementing this integration, enabling smoother information flow and enhanced utilization in construction projects. This paper presents a bibliometric analysis and provides a global perspective on research developments in the field of construction ERP and its integrations with BIM. It consolidates publication data from various databases (e.g., Scopus and Web of Science) as per geographical spread, authors, and number of publications as per the keyword search. It observes key research themes in each phase of the analysis. This paper concludes that although the existing literature has established the benefits of BIM and ERP integration, research is still in its nascent stage and identifies research gaps that can be taken up for further research.



Citation: Galagali, A.A.; Sandbhor, S.; Ruikar, K. A Survey on Enterprise Resource Planning and Building Information Modeling Integration: A Construction 4.0 Perspective.

Buildings **2024**, *14*, 3165. <https://doi.org/10.3390/buildings14103165>

Academic Editor: Ahmed Senouci

Received: 7 August 2024

Revised: 28 September 2024

Accepted: 2 October 2024

Published: 4 October 2024



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Keywords: building information modeling (BIM); enterprise resource planning (ERP); information systems (IS); integration; project management software (PMS)

1. Introduction

Information systems (ISs) in construction refer to managing, organizing, and analyzing information related to construction projects using technology and software. These systems enable construction companies to streamline their operations, improve communication and collaboration among team members, and make informed decisions based on real-time data.

A popular organizational IS is the Enterprise Resource Planning (ERP) system, which is used to integrate and manage various business operations, such as accounting, financial management, and supply chain management. In the construction industry, ERP systems are used to manage project costs, procurement, and inventory. The various capabilities of construction ERPs include project management involving programs of work, scheduling, requests for information, document management, workflow management, budget control, and production control. They also have financial capabilities for handling budgeting, procurement, quotation and estimation, tendering, invoicing, etc. Most of them also include Client Management (CM) and Human Resource Management (HRM) as well. Some globally popular ERP systems that have been specifically designed for construction are Acumatica, Viewpoint Vista, CMIC, NetSuite, and many more. Some ERP systems, like Acumatica, provide integration opportunities with PMS, like Procore, to integrate financial data [1].

In the phase of Industry 4.0, the construction industry is also gearing up toward the latest technological developments. Building Information Modeling (BIM), Common Data

Environment (CDE), Internet of Things (IoT) in construction, applications of Artificial Intelligence, Machine Learning, Deep Learning (AI/ML/DL) in construction, cloud computing, etc., have been popular areas of research in recent years. There is, however, a delay identified in the implementation of digitalization in construction. The reasons for this delay are chalked up to inadequate collaborations between stakeholders and their processes, insufficient knowledge transfer within organizations and unstandardized processes. This is further fueled by cultural and mindset issues that resist change to traditional practices [2]. The first step towards addressing this delay in the implementation of digitalization is to manage and organize data and information through the use of Information Systems (ISs) in daily operations.

Amongst the most dominant IS for construction project management is BIM, with the help of which, construction may be virtually planned before it is physically carried out in order to decrease uncertainty, increase safety, address issues, and assess possible effects [3]. BIM is often incorporated by Project Management Software (PMS), which is used to manage and track construction projects, including tasks, timelines, budgets, and resources. Some globally popular PMS are Procore construction management systems, Autodesk Build, BuilderTrend, CoConstruct, Oracle Primavera, and many more. Their capabilities vary and they need to be selected based on specific organizational needs.

Although BIM and ERP systems are developed for specific project and organizational roles, there are some common areas that both systems deal with, and they have the potential for effective integration. Both BIM and ERP are intended to solve the fragmentation of information within organizations. While ERP handles the operational domain, BIM handles the project information domain [4]. Thus, BIM can enable the management of project design information in the design phase, while ERP can execute operations correctly during the operational phase. Santos (2009) provides a comparative analysis of both BIM and ERP in terms of the evolution of both systems until 2009. He highlights that “BIM tools are information systems (IS), which contribute to enterprise resource planning (ERP)” [4,5]. Babič et al. provide the benefits of a BIM-CAD-ERP system in project monitoring and material flow management [6]. You and Wu proposed a novel BIM and ERP-linked framework called “Enterprise Integrated Data Platform” for interoperability between the two systems. Historical data, both from BIM and ERP, were collected on a unified big data platform that enabled knowledge discovery and decision-making [7]. Although research on BIM and ERP integration is in its nascent stage, a significant amount of research exists in the literature, explaining its benefits [8–12]. They have been highlighted as follows:

1. Finance Management
2. Accurate estimates for every item in the Bill of Quantities (BOQ) and the work required to complete the project can be obtained by combining BIM and ERP. This makes project resource planning more predictable and profitable.
3. Improved performance
4. An essential component of a construction project’s success is effective communication. Everybody involved in a project can have access to a single source of information due to the integration of BIM and ERP. These can be put into established procedures to ensure that the appropriate individuals receive real-time updates when needed.
5. Preventing contract conflicts
6. An integrated system aids in avoiding disputes over contracts with clients. To do this, a baseline estimate is created, and the forecast is then compared to the inventory, labor, raw material requirements, finances, and project budget.
7. Multiple project data simplification
8. Confusion in communication that may arise while managing data from multiple projects at a time can be avoided [3].

A good example of BIM-ERP integration can be seen with Acumatica, the cloud-based ERP system’s Construction edition with the Procore-built connector. It provides teams in the field and office the ability to share vendor, project, budget, commitment, change order, and project profitability data [1]. Integrated systems like these are seldom used in

developing countries due to economic factors, such as costs of implementation, which is a challenge for Small and Medium Enterprises (SMEs) [13]. The call for the digitalization of SMEs, especially during challenging times, such as the recent global COVID-19 pandemic, is also prevalent. Studies have also highlighted the difference between what researchers and digital developers recommend and the ground reality on construction sites. The development of real-world, cost-effective IS to support rebuilding and boost productivity is the need of the hour, especially in developing countries [14].

From the literature reviewed, it was concluded that even though both BIM and ERP systems are commonly adopted by the construction industry, the benefits that can be achieved by their integration have not yet been thoroughly investigated. A few gaps were also identified from the literature review. The benefits of BIM and ERP integration have been established without dwelling on the technicalities of software integration. There is also a dearth of case studies in existing research papers dealing with BIM and ERP integration.

Thus, a bibliometric survey was carried out with the objective to uncover the global spread of research interest and to identify prominent researchers and publication sources as well as emerging trends and themes around the research area of BIM and ERP integration. This research shall help to narrow down focus areas in this field as well as provide a base for further investigating the implementation of BIM-ERP integrated systems for efficient construction management.

2. Materials and Methods

Both construction ERP and BIM-ERP integrated systems were analyzed separately. The research was conducted in two phases, as illustrated in Figure 1.

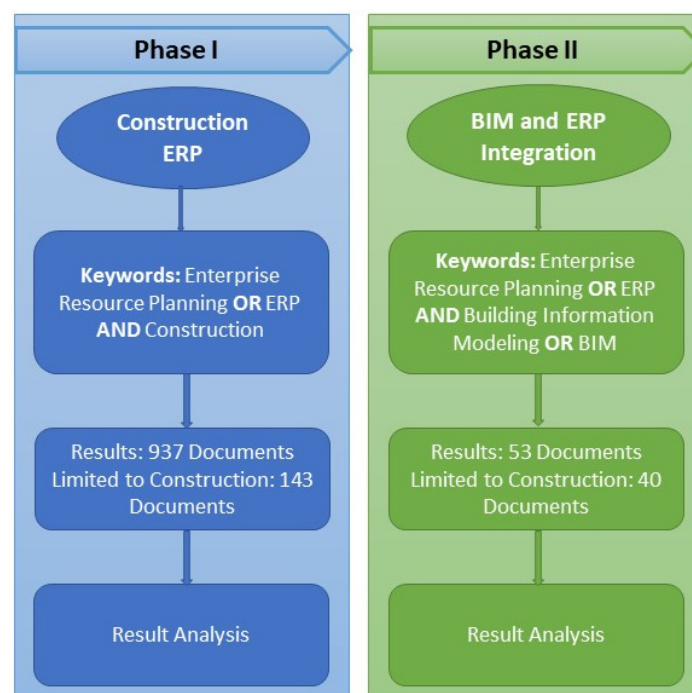


Figure 1. Phases of bibliometric survey with their respective methodology and results.

Phase I explores the current status of “Construction ERP”, while Phase II explores “BIM and ERP integration in construction management”. Each phase is divided into 6 sections, namely, documents by countries, authors, years, publication sources, keyword co-occurrences and phase summary. This was carried out to identify the global spread of research interest, prominent researchers, publication sources, and emerging themes around the research area, respectively.

In Section 5, finding keyword co-occurrences within the dataset was carried out using the software tool VOSviewer (version 1.6.19) (developed by Leiden University, Leiden, Netherlands). The tool helps to visualize the bibliometric network and is particularly useful in evolving areas of study, such as integrations of information systems like BIM and ERP [15].

The following sections thus provide an overall view of construction ERP and BIM-ERP integrated system implementation in line with the digital transformation of the construction industry under the concept of Construction 4.0.

In this study, Scopus and Web of Science (WoS) databases were used to survey the selected domain. A keyword search was conducted for both the phases of the study. Various keyword combinations were used for the same, such as “BIM”, “Building Information Modeling”, “Enterprise Resource Planning”, “ERP”, etc., including various Booleans, such as “AND”, “OR”, and “AND NOT.” As the relevant articles from WoS formed a subset of those in Scopus, the latter was primarily used. It was evident from the initial keyword searches that there is currently no bibliometric survey in the selected research area. Thus, this paper is an attempt to fill that gap.

3. Phase I: Bibliometric Survey of Construction ERP

Scopus “Article Title/Abstract/Keyword” search was used with the combination of keywords (“Enterprise Resource Planning” OR “ERP”) AND “construction”. As seen in Figure 1, this led to 937 publications ranging from the year 1953 to 2023, which included research from various fields outside of the construction industry like energy, medicine, and many more. Since this phase was exploring the use of ERP specific to the construction industry, relevant filters specific to engineering and the construction sector were applied. This led to 143 results ranging from the year 2000 to 2023. Overall, 137 of the 143 publications were from a total of 35 different countries, and 6 out of 143 were from undefined countries. The analysis was further refined based on relevancy to identifying the major countries, authors, publications, and keywords prevalent in the research.

3.1. Documents by Countries (Phase I–Construction ERP)

The USA has produced the highest number of 22 publications in the research area of construction ERP, followed by China with 20 publications, and the UK with 13 publications. The remaining countries, as shown in Figure 2, from South Korea to Turkey, have all produced more than three publications each. Apart from the ones displayed, the countries of Denmark, Finland, Greece, the Netherlands, Saudi Arabia, Slovakia, South Africa, and Sri Lanka have three publications each; the countries of India, Iran, Italy, New Zealand, the Russian Federation, and Spain have two publications each; and Cameroon, Colombia, Norway, Peru, Portugal, Slovenia, and Ukraine have one publication each. Six publications, however, are linked to undefined countries.

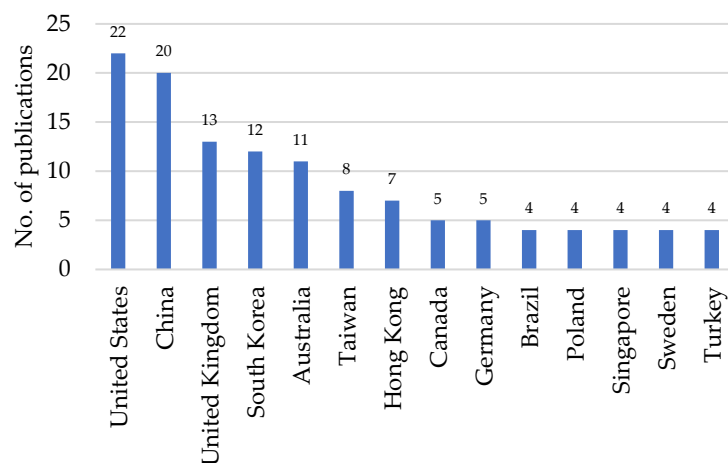


Figure 2. Documents by countries (Phase I–Construction ERP).

3.2. Documents by Authors (Phase I–Construction ERP)

The top author in the area of construction ERP was identified as M.J. Skibniewski, who has authored and co-authored a total of eight publications in the domain area. Skibniewski's prominent works include exploring the state of construction ERP systems in 2007 [1], identifying success factors and performance indicators of construction ERP systems [2,3] and developing a decision-making model for small- and medium-scale enterprises [4], among many others. Other authors with a minimum of two publications each can be identified in Figure 3.

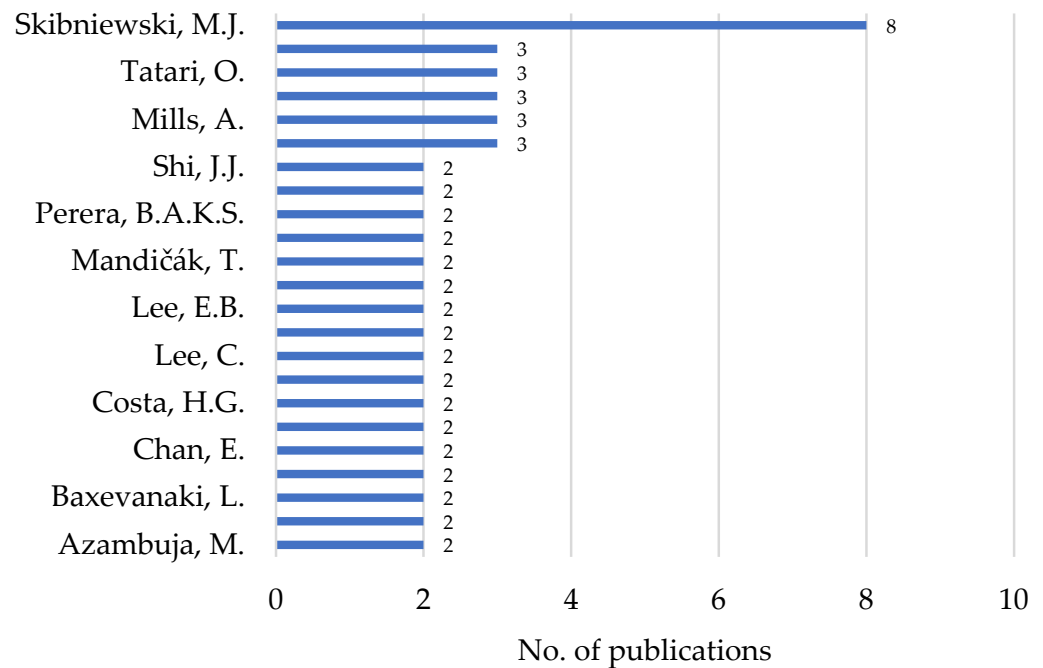


Figure 3. Documents by authors (Phase I–Construction ERP).

3.3. Documents by Year (Phase I–Construction ERP)

As shown in Figure 4, research on construction ERP rose from the year 2000 until it hit its peak in 2009 with 15 publications. Since then, the publication trend has been fluctuating, but on an overall decline, with two papers published in 2023.

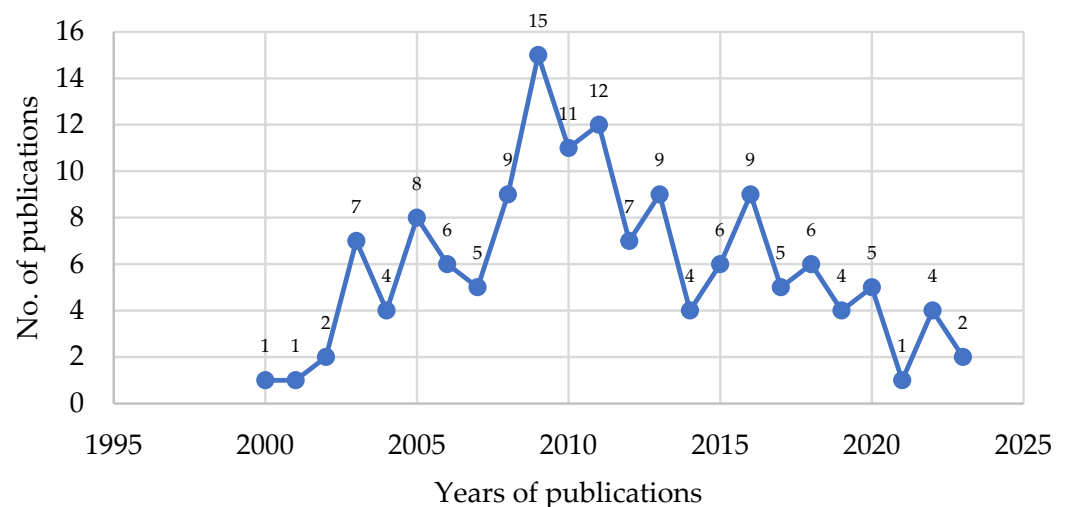


Figure 4. Documents by year (Phase I–Construction ERP).

3.4. Documents Per Source (Phase I–Construction ERP)

The most prominent publication sources in the research area of construction ERP have been identified, as shown in Figures 5 and 6. The trends in publications, however, are split into 2 time periods, namely “2000–2018” and “2019–2023,” since the prominent publication sources are different for both periods. The second period split has also been intentionally provided to indicate the trend in the last 5 years.

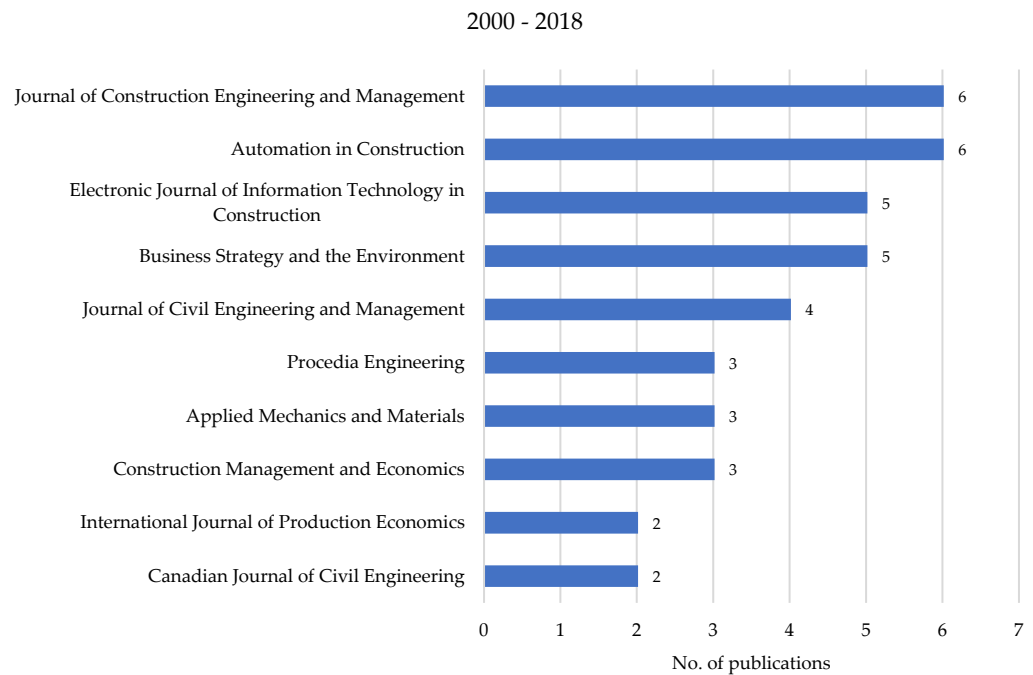


Figure 5. Prominent publication sources in the years 2000–2018 (Phase I–Construction ERP).

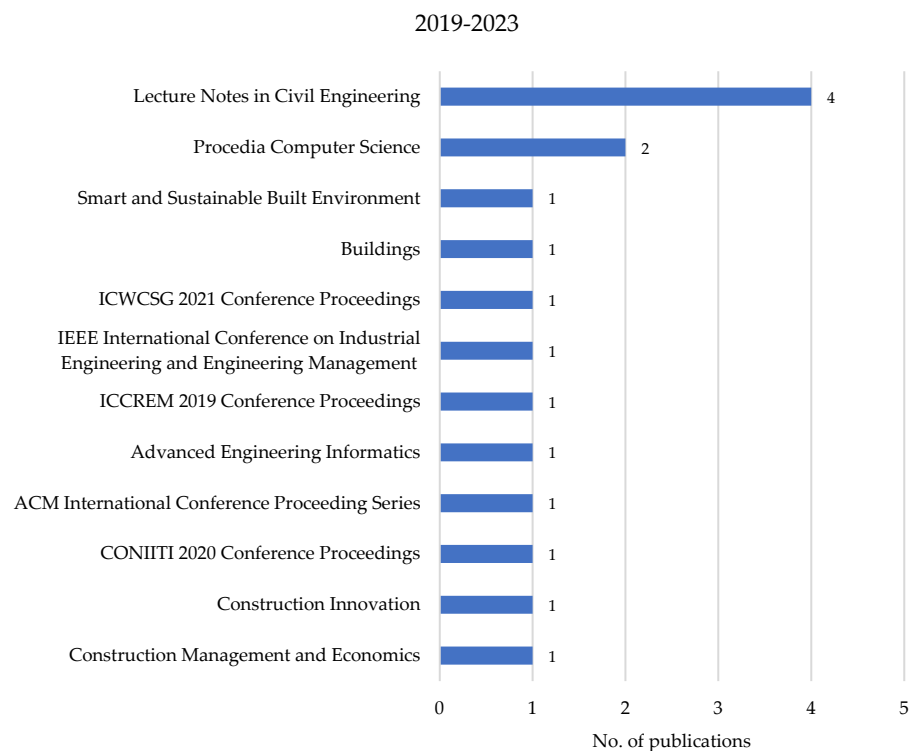


Figure 6. Prominent publication sources in the years 2019–2023 (Phase I–Construction ERP).

As seen from Figure 5, “Journal of Construction Engineering and Management” and “Automation in Construction” were the top publication sources for construction ERP, with a total of six publications each in the time period of 2000 to 2018. Other prominent sources were the “Electronic Journal of Information Technology in Construction”, “Business Strategy and the Environment”, and the “Journal of Civil Engineering and Management”, with no less than four publications each in that time period.

As shown in Figure 6, post-2019, different publication sources seem to be taking the lead in publications in the research domain of construction ERP, namely “Lecture Notes in Civil Engineering”, with four publications since 2020, followed by “Procedia Computer Science,” “Smart and Sustainable Built Environment”, and “Buildings”, with one publication each since 2022.

3.5. Keyword Occurrence Analysis Using VOSviewer (Phase I–Construction ERP)

All keyword occurrences were analyzed from the Scopus dataset for construction ERP to obtain the network, as shown in Figure 7. Keywords that frequently occur together within the dataset are grouped into a common color cluster, indicating a shared focus. The network displayed five clusters of co-occurring keywords in the colors red, green, blue, yellow, and purple. The most prominent keyword co-occurrences with their respective link strengths are provided in Table 1. The higher the keyword co-occurrence within the dataset, the higher the total link strength. These keywords appear in bigger font sizes and, thus, can be easily visualized in the network. As seen in Table 1, the keywords that co-occurred most frequently were “ERP” and “Construction industry”, owing to the focus area being construction ERP itself. It was notably followed by “cost management”, “collaborative working”, “information management”, “BIM”, and eventually, “Procurement strategy”, “System integration”, and “Sustainability”, indicating less explored but upcoming opportunities for further research in ERP and BIM integration.

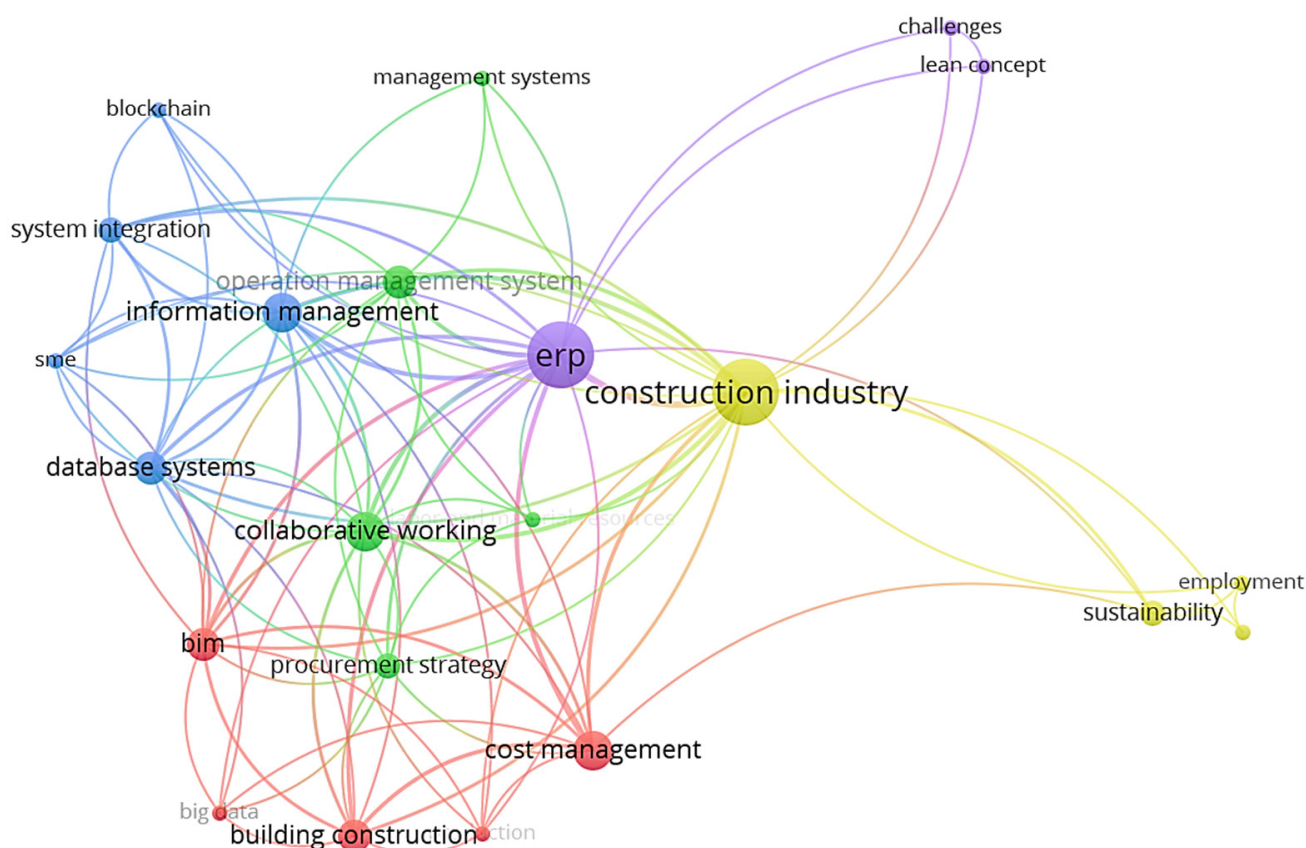


Figure 7. Keyword occurrences (Phase I–Construction ERP).

The purple cluster, as shown in Figure 8, focuses on links with keyword “erp” and its co-occurrence with other keywords in the dataset. The cluster illustrates ERP’s links with “lean concepts” as well as “challenges” in its implementation. It also displays links with the other clusters, with close associations with various concepts of the construction industry such as “information management”, “BIM”, “cost management”, “sustainability”, and many others.

Table 1. Most prominent keyword occurrences for construction ERP (Phase I–Construction ERP).

Sr.no	Keywords	Occurrences	Total Link Strength
1	Construction industry	9	42
2	ERP	9	45
3	Cost management	4	20
4	Collaborative working	4	24
5	Information management	4	24
6	BIM	3	21
7	Building construction	3	17
8	Operations management systems	3	17
9	Database systems	3	19
10	Procurement strategy	2	11
11	System integration	2	13
12	Sustainability	2	6

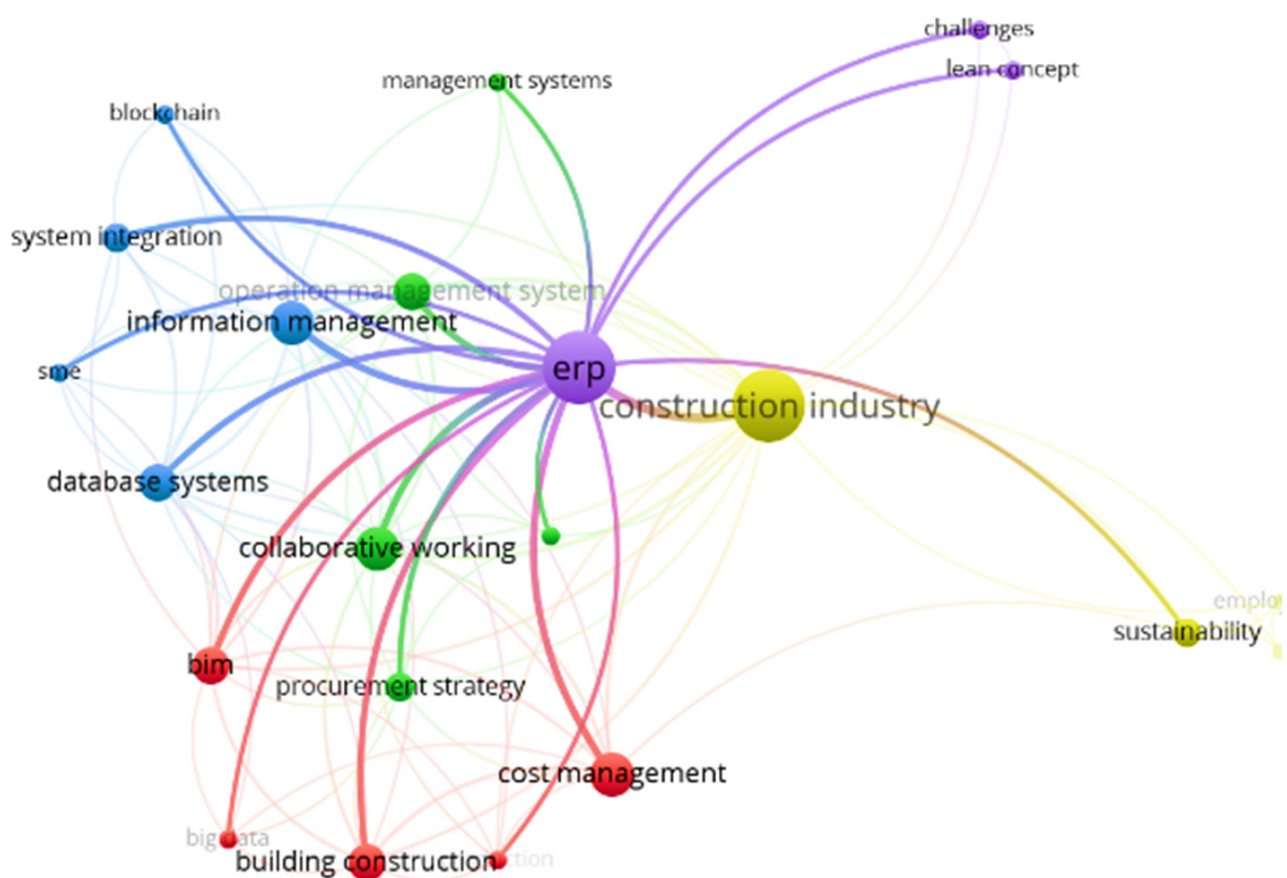


Figure 8. Purple cluster with a focus on ERP (Phase I–Construction ERP).

Similar keyword co-occurrences were observed with the red cluster with a focus on the keyword “BIM”, as shown in Figure 9. This cluster highlighted the “collaborative working” nature of their “system integration”. The link to “sme” (Small and Medium Enterprises) also illustrates that BIM and ERP integration for SMEs is being explored by researchers.

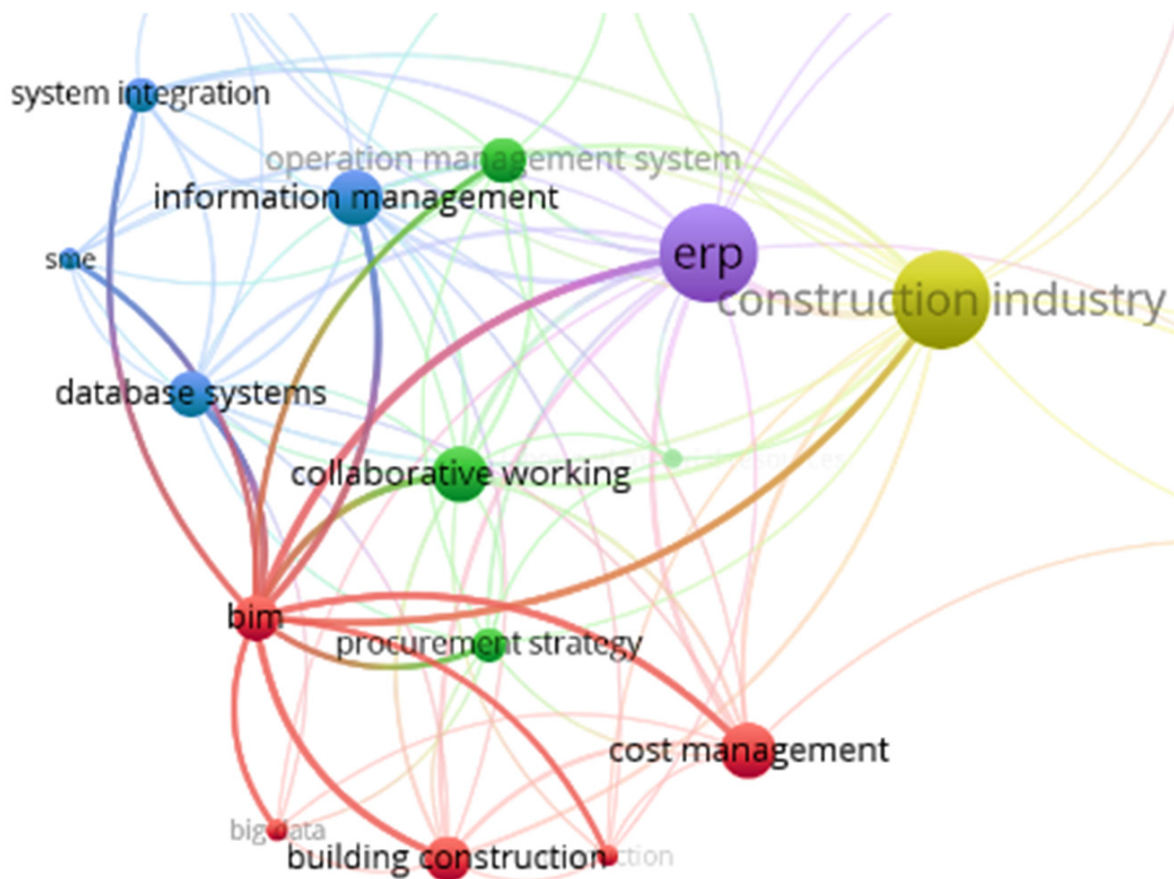


Figure 9. Red cluster with a focus on BIM (Phase I–Construction ERP).

3.6. Summary of Bibliometric Survey for Phase I–Construction ERP

Table 2 summarizes the bibliometric survey results of Phase I–Construction ERP. It includes the total number of documents published, the top 3 countries with the maximum number of publications, the top authors with highly cited articles, and the top publication sources in Phase I.

Table 2. Summary of bibliometric survey for Phase I–Construction ERP.

	Phase I
Total documents	143
Top 3 countries	US, China, UK
Top authors	Skibniewski, Chan, Mills
Top journals	1. <i>Automation in Construction</i> 2. <i>Journal of Construction Engineering and Management</i>

A substantial global spread of research interest in construction ERP can be seen, with a maximum of 22 publications from leading countries. M.J. Skibniewski, who has authored and co-authored a total of eight publications in the domain area, has been identified as the top author in the domain. Beyond its peak in 2009 with 15 publications, the publication trend has been fluctuating but on an overall decline, with only 2 papers published so far in

2023. Strong keyword links can be interpreted as showing strong relationships between ERP, BIM, collaborative working, and system integration.

4. Phase II: BIM and ERP Integration

Scopus “Article Title/Abstract/Keyword” search was used with the combination of keywords (“Enterprise Resource Planning” OR “ERP”) AND (“Building Information Modeling” OR “BIM”). As seen in Figure 1, this led to 53 publications from various fields outside of the construction industry, including medical, social sciences, etc. Since this research explores the use of BIM, ERP and their integration specific to the construction industry, relevant filters specific to engineering and the construction sector were applied. This led to 40 publications ranging from the years 2005 to 2023. Overall, 33 of the 40 publications are from a total of 20 different countries, and 7 out of 40 are from undefined countries. The analysis is further refined based on relevancy to identifying the major countries, authors, publications, and keywords prevalent in the research.

4.1. Documents by Countries (Phase II–BIM and ERP integration)

China and the USA are leading in publications in the domain area of BIM and ERP integration, with four publications each. They are followed by Canada, Germany, and the UK, with three publications each. As seen in Figure 10, the countries of Australia, the Czech Republic, Finland, India, Poland, Slovenia, Taiwan, and Turkey have all produced two publications each. While the countries of Brazil, Kazakhstan, Slovakia, South Korea, Sri Lanka, and the United Arab Emirates all have one publication to their names, 7 out of the total 40 publications are associated with undefined countries.

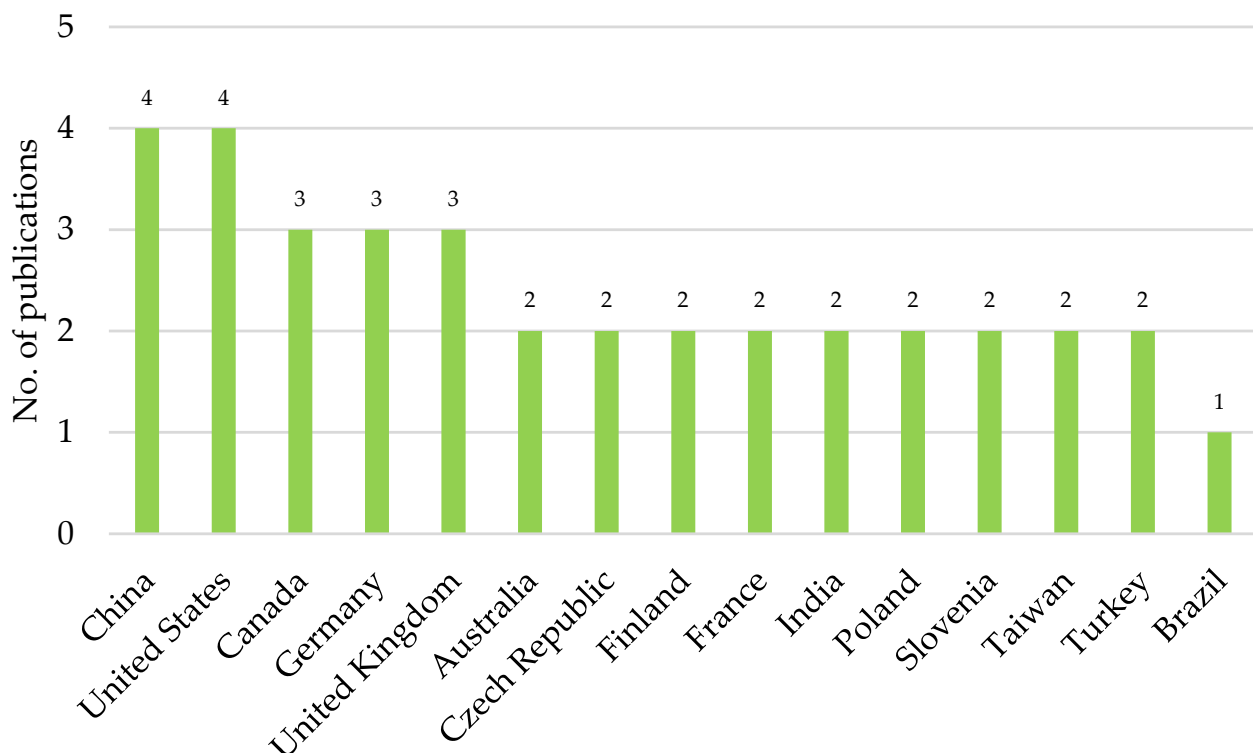


Figure 10. Documents by countries (Phase II–BIM and ERP integration).

4.2. Documents by Authors (Phase II–BIM and ERP integration)

Since the number of publications in the domain of BIM and ERP integration was limited, determining the number of publications per author to find prominent contributors could not be justified. Hence, the most cited articles in the research area of BIM and ERP integration were investigated to identify prominent contributors in the domain. The results

are shown in Table 3. Prominent works of these authors in the domain of BIM and ERP integrations include research to integrate mass-produced prefabrication processes with construction site activities [6], exploration through case studies of improved information flow with BIM and ERP integration [8], comparisons of the most popular modes of Information Management Systems [12], IoT-based material management systems with BIM and ERP integration [9], proposals of a conceptual framework for integrating BIM, ERP and process management systems to improve supply management, enabling remote project delivery and automation in project processes in the construction industry [16,17].

Table 3. Authors of most-cited articles on BIM and ERP integration.

Sr.no	Authors	Title	Year	Source Title	Cited by
1	Babič N.C.; Podbreznik P.; Rebolj D. [6]	Integrating resource production and construction using BIM	2010	<i>Automation in Construction</i>	128
2	Demian P.; Walters D. [8]	The advantages of information management through building information modelling	2014	<i>Construction Management and Economics</i>	110
3	Wang M.; Altaf M.S.; Al-Hussein M.; Ma Y. [9]	Framework for an IoT-based shop floor material management system for panelized homebuilding	2020	<i>International Journal of Construction Management</i>	45
4	Sarkar D.; Pandya K.; Dave B.; Jha K.N.; Dhaneshwar D. [17]	Development of an integrated BIM-ERP-IoT module for construction projects in Ahmedabad	2022	<i>Innovative Infrastructure Solutions</i>	8
5	Shokri S.; Safa M.; Haas C.T.; Haas R.C.G. [16]	A conceptual framework to improve information and process management in the execution of capital projects	2011	<i>Proceedings, Annual Conference—Canadian Society for Civil Engineering</i>	7

4.3. Documents by Year (Phase II–BIM and ERP integration)

As shown in Figure 11, the publication in the area of BIM and ERP integration has been gaining interest since 2005, with the highest publication being in the year 2020, with five publications. It is evident that this area of research is not the most popular but is gaining attention, as seen from an upward linear trendline.

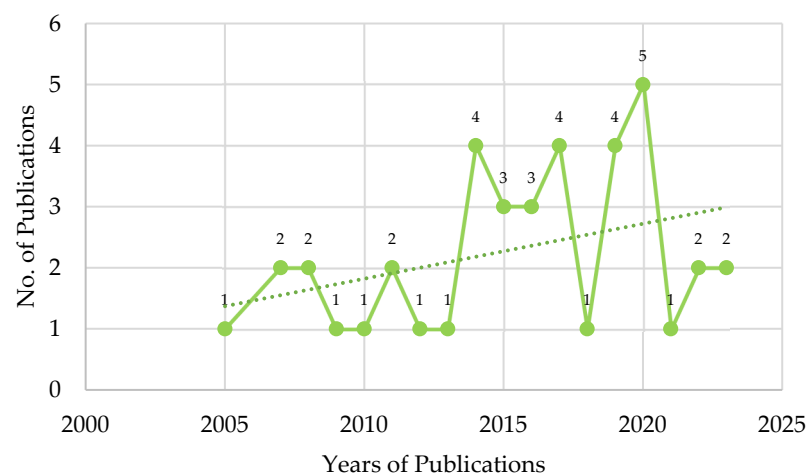


Figure 11. Documents by year (Phase II–BIM and ERP integration).

4.4. Documents by Source (Phase II–BIM and ERP integration)

The top 10 prominent publication sources in the research area of BIM and ERP integration are shown in Figure 12. “Automation in Construction”, “Lecture Notes in Civil Engineering”, and “Betonwerk und Fertigteil-Technik (Concrete Plant and Precast Technology)” are the leading publications in the research area, with three publications each. These are followed by “Advanced Material Research” and “Buildings”, with two publications each.

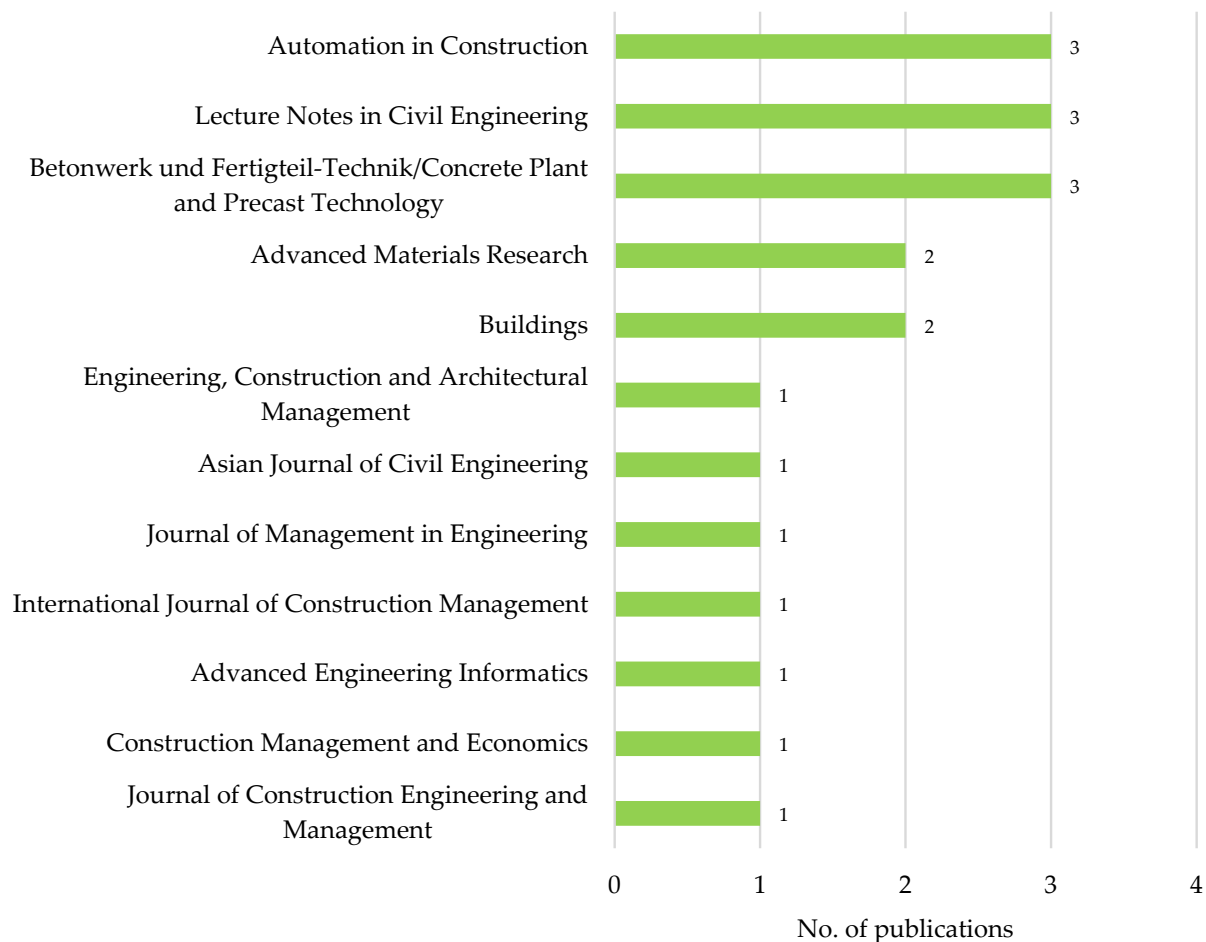


Figure 12. Documents by source (Phase II–BIM and ERP integration).

4.5. Keyword Occurrence Analysis Using VOSviewer (Phase II–BIM and ERP integration)

Upon analysis of all keyword co-occurrences in the articles obtained from the Scopus dataset under the domain of BIM and ERP integration, the network obtained is presented in Figure 13. The network displayed three clusters of co-occurring keywords in the colors red, green, and blue. The network displays strong links between ERP systems and BIM in construction management. It also makes it clear that BIM and ERP integrations are usually linked with the concept of sustainability. The available literature also deals with identifying critical success factors for BIM and ERP integration implementations.

The most prominent keyword co-occurrences with their respective link strengths are provided in Table 4. The keywords that co-occurred most frequently were “BIM” and “ERP” owing to the focus area being ERP and BIM integrated systems. It is notably followed by “Information Systems”, “Construction Management”, “Integration”, and eventually “Sustainability” and “Critical Success Factors”, indicating less explored but upcoming opportunities for further research into the sustainability of ERP and BIM integration and determining critical success factors in their implementation.

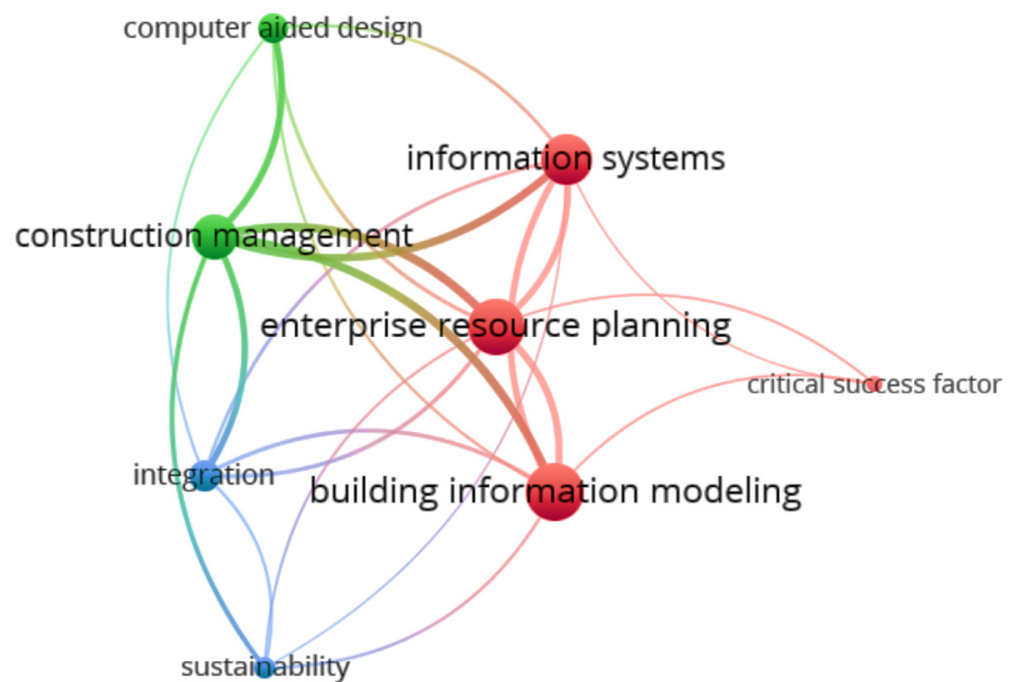


Figure 13. Keyword occurrences (Phase II–BIM and ERP integration).

Table 4. Most prominent keyword occurrences for BIM and ERP integration.

Sr.no	Keywords	Occurrences	Total Link Strength
1	Building Information Modeling	25	60
2	Enterprise Resource Planning	24	62
3	Information System	20	51
4	Construction Management	15	42
5	Integration	5	27
6	Computer Aided Design	6	19
7	Sustainability	3	10
8	Critical Success Factor	2	5

4.6. Summary of Bibliometric Survey for Phase II–BIM and ERP integration

Table 5 summarizes the bibliometric survey results of Phase II: BIM and ERP integration. It includes the total number of documents published, the top three countries with the maximum number of publications, the top authors with highly cited articles, and the top publication sources.

Table 5. Summary of bibliometric survey for Phase II–BIM and ERP integration.

Phase II	
Total documents	40
Top 3 countries	China, US, Canada
Top authors	Babič N.C. [6]; Podbreznik P. [6]; Rebolj D, Demian P. [6]; Walters D. [8]
Top journals	1. <i>Automation in construction</i> , 2. <i>Betonwerk und Fertigteil-Technik</i> 3. <i>Lecture Notes in Civil Engineering</i>

Research on BIM and ERP integration is still in its nascent stage, with a maximum of four publications from leading countries. Publication numbers per source have been low but consistent, with at least one article being published per year, and a maximum of two per year by leading publication sources. It is evident that this area of research is not the most popular but is gaining attention. Strong keyword links make it clear that BIM and ERP integrations are usually linked with the concept of sustainability.

5. Discussion

The construction industry, especially in developing countries, has been notoriously behind in adopting the digital transformation possible in this age of Construction 4.0. The existing literature has expounded the reasons, which range from the absence of government mandates to mindset issues and everything in between. One way to make it easier for construction firms to open up to the adoption of technological advancements is to provide in-house solutions using existing information systems. Both BIM and ERP systems are popular among construction firms, with the former handling project information and the latter handling business operations. Both perform distinct functions, with a few but significant overlaps. Exploring their integration for the benefits of better financial management, improved project performance, contractual conflict prevention, and multiple project data simplification, to name a few, is a worthwhile undertaking.

This paper forms part of the research that focuses on “BIM-ERP integrated systems for efficient construction management”. Thus, a bibliometric survey was conducted at the early stages of this research to uncover the global spread of research interest and identify prominent researchers and publication sources, as well as emerging trends and themes around the research area of BIM and ERP integration. The survey was conducted in two phases. Phase I focused on research on ERP systems specific to the construction industry, while Phase II focused on research on BIM and ERP integrated systems.

In Phase I of this research, it is evident that the USA, China, and the UK are the top three countries leading in research on construction ERP. With China as the exception, it is observed that research in this field is mostly undertaken in developed countries. This can be attributed to the prevalence of large-scale enterprises in these countries with more scope for research and development. Mirosław J. Skibniewski has been identified as a prominent author in the research area of construction ERP, followed by Caroline T W Chan, Anthony Mills, and Omer Tatari. Notable focus areas from these prominent authors include exploring the state of construction ERP systems, identifying success factors and performance indicators for construction ERP systems, and developing a decision-making model for SMEs. The research on construction ERP began in 2000, reached its peak with 15 publications in 2009, and has since declined to just 2 publications in 2023. The decline could be attributed to the development and prevalence of various construction ERP software in the market, like Acumatica, Viewpoint Vista, CMIC, NetSuite, and many more. It was observed that prominent publication sources can differ depending on the time period. While “Automation in Construction” and “Journal of Construction Engineering and Management” were the top publication sources for construction ERP research from 2000 to 2018, “Lecture Notes in Civil Engineering” and “Procedia Computer Science” were the top publication sources post-2019. The co-occurrence network of keywords facilitated the identification of new research focus areas, specifically “procurement strategy” and “sustainability”, in relation to ERP systems in the construction sector. Phase I also concluded that there were some gaps in the area of research on construction ERP. There is no literature exploring the current state of construction ERPs post-2007. Existing studies limited data collection to large-scale contracting firms, and only a few addressed ERP systems for SMEs prevalent in developing countries. Most research primarily relied on qualitative analysis. There is a significant lack of research into practical software integrations with ERP systems for construction. Because of the various technological advancements associated with Construction 4.0, there is a need to explore the many possibilities for improvements in current construction ERP systems.

Phase II of this research revealed that the USA and China once again topped the list of countries conducting research on BIM and ERP integrations. The disparity between contributions from developing and developed countries, however, was far lower than that in Phase I. This can be attributed to the increased popularity of BIM in developing countries. It can also be interpreted as research on BIM and ERP integration being at a nascent stage. An observation of a total of 40 publications in the period 2005–2023 further reinforced this point. It was evident that although the number of publications was few, publications on the research topic have been consistent from 2005 to 2023, with “Automation in Construction” and “Lecture Notes in Civil Engineering” being top publication sources. This consistency indicated that researchers are interested in exploring information systems integrations in the construction sector. Work from prominent authors in the field included research to integrate mass-produced prefabrication processes with construction site activities, comparisons of the most popular modes of Information Management Systems and developing conceptual frameworks for BIM and ERP integrated platforms for business improvement. The keyword co-occurrence network established that BIM and ERP integrations are being viewed through the lenses of sustainability. It also showed that identifying critical success factors for BIM and ERP integration was among the focus areas in this field. Phase II concluded there exists a scope for real-world case studies on BIM and ERP integration that should be verified by practical exercises on a sizable scale.

Most cited works in this domain of both construction ERP and its integration with BIM pointed towards interdisciplinary opportunities in research, combining construction management, information technology, and systems engineering. Researchers are exploring specific research areas such as improving information exchange, enhancing supply chain management, and automating processes in construction. These areas integrated with the technologies of Construction 4.0, like IoT, AI/ML/DL, etc., were identified as ripe for detailed study that could yield significant industry improvements.

In a world with no shortage of data being generated on construction sites and businesses, the integration of powerful systems like BIM and ERP can be a robust step towards realizing the benefits of Construction 4.0.

Author Contributions: Conceptualization, A.A.G. and S.S.; methodology, A.A.G., S.S. and K.R.; software, A.A.G.; validation, S.S. and K.R.; formal analysis, A.A.G.; investigation, A.A.G.; resources, S.S. and K.R.; data curation, A.A.G.; writing—original draft preparation, A.A.G.; writing—review and editing, S.S. and K.R.; visualization, A.A.G.; supervision, S.S. and K.R.; project administration, S.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The datasets presented in this article are not readily available because the data are part of an ongoing study. Requests to access the datasets should be directed to the corresponding author.

Acknowledgments: I am deeply grateful to my guide, Sayali Sandbhor, for her invaluable support and insightful feedback that have been instrumental in shaping this research paper. I also extend my appreciation to my co-guide, Kirti Ruikar, for her valuable inputs and constructive suggestions that have significantly enriched the content of this paper. Special thanks are due to my fellow research scholars, Shalaka Hire and Pradnya Desai, for their meaningful discussions that enriched the depth of my understanding. Finally, a heartfelt appreciation to my family for their unwavering support and understanding during this academic journey. Their encouragement has been a constant source of inspiration. I am fortunate to have such a dedicated academic support system and a loving family, and I thank them for their contributions to my research endeavors.

Conflicts of Interest: The authors declare no conflicts of interest.

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